

Questions by topic: Forest resources & deforestation - Set 1 (55 marks)

Answer all questions in the spaces provided.

01 . 1

Describe the resources obtained from forests. Give **one** example for each category.

[5 marks]

Timber

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Fuel

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Food

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Fibres

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Medicines

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01 . 2

Explain why forests are an important carbon reservoir and how they can be used to counteract climate change.

[3 marks]

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01 . 3

Explain how forests regulate the hydrological cycle and local climate.

[2 marks]

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10

0 2

Table 1 compares an indigenous mixed broadleaf woodland with a Sitka spruce plantation of the same area in Scotland.

Table 1

Feature	Mixed indigenous woodland	Sitka spruce plantation
Tree species	12	1
Age structure	All ages, dead wood present	All planted in the same year
Planting density	Natural, variable	Close, 2 m spacing
Timber yield / $\text{m}^3 \text{ ha}^{-1} \text{ yr}^{-1}$	4	16
Bird species	38	9
Ground flora species	60	4

0 2

. 1

Use Table 1 to explain why the plantation has a much higher timber yield but far lower biodiversity.

[6 marks]

0 2 . 2

Explain why commercial plantations are felled before the trees reach full size.

[2 marks]

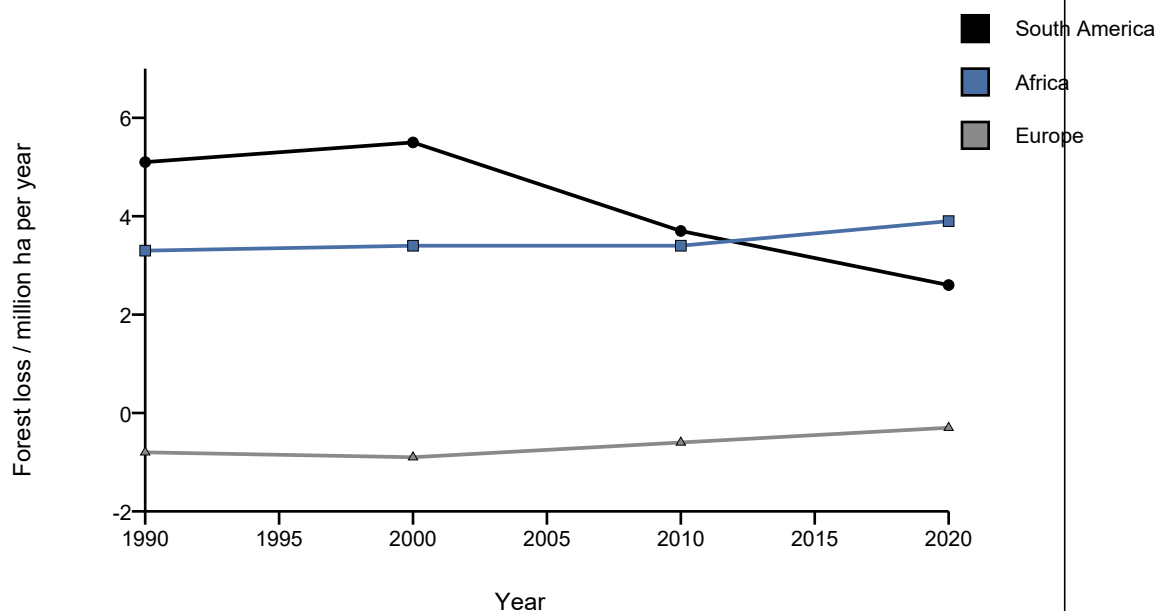
0 2 . 3

Explain why there is little demand today for coppiced or pollarded wood.

[2 marks]**10**

0 3

Figure 1 shows the area of forest lost each year in three regions between 1990 and 2020.

Figure 1

0 3 . 1

Describe the trends shown in Figure 1.

[3 marks]

03 . 2

Explain the main causes of forest loss in South America and Africa.

[4 marks]

03 . 3

Explain why roads and railways cause more forest loss than the area they occupy.

[2 marks]

03 . 4

Explain why softwood from Scandinavia and Canada is generally produced more sustainably than tropical hardwood.

[1 mark]

10

0 4 . 1

Explain the effects of deforestation on soil.

[4 marks]

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0 4 . 2

Explain how deforestation affects hydrology and climate.

[4 marks]

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0 4 . 3

Give **two** ecological impacts of deforestation other than the direct loss of species.

[2 marks]

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0 5

The Rio Bravo Conservation and Management Area in Belize exploits mahogany under a sustainable management plan while conserving a rainforest of over 200 tree, 400 bird and 70 mammal species.

0 5

1

Describe **five** features of the timber harvesting at Rio Bravo that make it sustainable.

[5 marks]

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0 5

2

Explain why involving local communities is essential to the success of the reserve.

[3 marks]

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0 5

3

Explain why selective logging is better for wildlife than clear-felling, and give **one** disadvantage.

[2 marks]

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10

06 . 1

Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

- A Close planting of a single non-indigenous species ☐
- B Mixed indigenous species with a mixed age structure ☐
- C Clear-felling on a 40-year rotation ☐
- D Removal of dead wood ☐

06 . 2

Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

- 1
- 2
- 3
- 4

5

END OF QUESTIONS